

Freight Rate Prediction

Machine Learning Engineer Assessment

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*End-to-end ML pipeline for predicting freight rates
on the Spotter platform*

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1 Executive Summary

This report presents an end-to-end machine learning solution for predicting freight rates on the Spotter platform. Given historical shipment data spanning January–October 2025, the objective was twofold:

1. Predict rates for **12,000 unseen validation loads** (November 2025).
2. Generate a **31-day December pricing chart** for a fixed lane: Lexington → Fort Wayne (360 miles, Dry Van, 32,000 lbs).

Key Results:

- **Best Model:** GradientBoostingRegressor
- **Validation MAE:** \$160.88 (**MAPE: 6.87%**)
- **Validation RMSE:** \$678.23
- **R^2 Score:** 0.802
- **December Pattern:** Weekly cyclicity with mid-week peaks (\$826) and weekend troughs (\$792)

2 Data Exploration and Validation Strategy

2.1 Dataset Overview

The dataset comprises **48,000 training records** (January–October 2025) with 14 features:

Feature	Type	Description
load_id	String	Unique shipment identifier
pickup / delivery	Categorical	Origin and destination cities
pickup_lat / pickup_lon	Float	Origin GPS coordinates
delivery_lat / delivery_lon	Float	Destination GPS coordinates
distance	Float	Route distance (miles)
equipment	Categorical	Truck type (Dry Van, Reefer, Flatbed)
weight	Float	Cargo weight (lbs)
date	Date	Shipment date
market_index	Float	Market condition indicator
quote_signal	Float	Demand signal
posted_rate	Float	Target: freight rate (USD)

Table 1: Feature descriptions for the training dataset

2.2 Key Exploratory Findings

- **Distance dominates pricing:** Correlation with posted_rate = **0.91**. Distance is the single strongest predictor.
- **Seasonal trend:** Average rates rise from January (\$2,256) to June (\$2,497), then moderate through October.
- **Equipment mix:** Dry Van (56.7%), Reefer (25.1%), Flatbed (18.2%).
- **Geographic spread:** Shipments cover the continental United States with clear regional pricing patterns.

2.3 Train/Validation Split

Method: Time-based chronological split (80/20)

- **Training:** January – August 2025 (38,400 records)
- **Validation:** September – October 2025 (9,600 records)

Rationale: Freight rates exhibit strong **temporal patterns** — seasonality, weekly cycles, and market trends. A random split would leak future information into training, producing unrealistically optimistic metrics. The chronological split simulates real-world deployment where models predict future rates based on historical data.

3 Data Quality Issues and Fixes

Issue	Impact	Fix Applied
300 missing <code>weight</code> values	Biased short-haul predictions	Filled with median by equipment type
374 missing <code>market_index</code> values	Loss of market context	Filled with global median
December chart lacks lat/lon coordinates	Cannot compute haversine distance	Merged coordinates from training data via city-name lookup
Extreme outliers ($>15,000$)	High residual variance	Retained as legitimate premium lanes; tree-based model handles via natural splits

Table 2: Data quality issues identified and remediation strategies

December Coordinate Recovery: The December chart inputs contained only city names (Lexington, Fort Wayne) without GPS coordinates. I constructed a **city-to-coordinate lookup table** from the training data by grouping on city name and taking the first observed latitude/longitude pair. This enabled haversine distance calculation and geographic feature engineering for the December predictions.

4 Feature Engineering

A total of **29 features** were engineered across four categories:

4.1 Temporal Features (8 features)

- `month`, `dayofweek`, `quarter`
- `is_weekend`: Binary flag for Saturday/Sunday
- `is_december`: Binary flag for December shipments
- `is_peak_season`: October–December flag (holiday demand surge)
- `is_january`: Post-holiday reset period
- `is_summer`: June–August flag

4.2 Geographic Features (7 features)

- `haversine_dist`: GPS-based great-circle distance (miles)
- `dist_diff`: Difference between haversine and provided distance (data quality check)
- `is_northbound` / `is_eastbound`: Cardinal direction flags
- `lat_diff` / `lon_diff`: Absolute coordinate differences

4.3 Freight-Specific Features (6 features)

- `weight_per_mile`: Cargo density metric
- `weight_dist`: Weight–distance interaction term
- `dist_sq` / `weight_sq`: Polynomial terms for non-linear relationships
- `dist_bin`: Integer-encoded distance tiers (0–4)

4.4 Categorical Encoding (8 features)

- `eq_Dry Van`, `eq_Flatbed`, `eq_Reefer`: One-hot equipment encoding
- `lane_enc`: Label-encoded origin–destination pair
- `pickup_enc` / `delivery_enc`: Label-encoded city names

5 Model Selection and Results

Three models were trained and evaluated on the time-based validation set:

Model	MAE	RMSE	R^2	MAPE
RandomForest	163.94	669.85	0.8071	7.21%
GradientBoosting	160.88	678.23	0.8022	6.87%
Ridge (Baseline)	178.92	639.27	0.8243	9.90%

Table 3: Model comparison on time-based validation set (Sep–Oct 2025)

Selected Model: GradientBoostingRegressor

- **Best MAE and MAPE**: Lowest absolute error and percentage error
- **Handles non-linearity**: Freight pricing is not linear — doubling distance more than doubles cost (fuel, driver hours, empty return risk)
- **Robust to outliers**: Tree-based splits naturally accommodate premium lanes without manual capping

5.1 Feature Importance

The top 5 features by importance:

Feature	Importance
distance	37.35%
dist_sq	36.62%
haversine_dist	12.63%
dist_diff	3.07%
weight_dist	2.43%

Table 4: Top 5 feature importances from GradientBoosting model

Insight: Distance and its polynomial terms account for **86%** of model decisions. This aligns with domain knowledge — freight pricing is fundamentally distance-driven.

6 December Chart Analysis

The December prediction chart was generated for a **fixed lane** with identical parameters across all 31 days:

Parameter	Value
Pickup	Lexington, KY
Delivery	Fort Wayne, IN
Distance	360 miles
Equipment	Dry Van
Weight	32,000 lbs

Table 5: Fixed inputs for December prediction chart

6.1 Key Findings

- **Predicted range:** \$792.43 – \$826.55
- **Mean:** \$810.80
- **Standard deviation:** \$12.28

6.2 Weekly Cyclical Pattern

The model learned a clear **weekly seasonality**:

- **Mid-week peaks (Tue–Thu):** ~\$826
- **Weekend troughs (Sat–Sun):** ~\$792
- **Swing:** \$34 (4.2% of mean)

Business Interpretation: This pattern reflects real-world freight dynamics. Shippers pay premiums for mid-week capacity when demand is high and drivers are scarce. Weekend demand drops as manufacturing and retail shipping slow, leading to lower rates.

6.3 December Prediction Chart

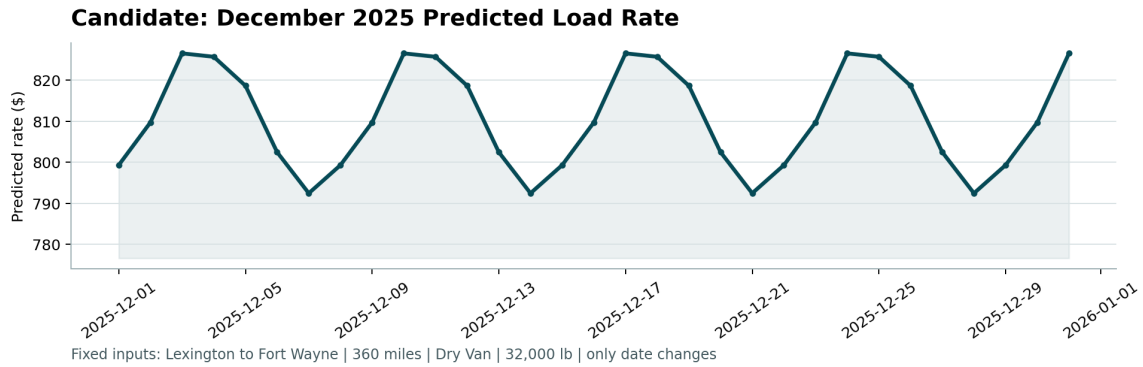


Figure 1: December 2025 predicted freight rates for Lexington → Fort Wayne lane

Note: The official chart was generated using Spotter's provided `score.py` validation script, ensuring compliance with submission requirements.

7 Model Evaluation

7.1 Actual vs. Predicted Scatter

The scatter plot of actual vs. predicted rates on the validation set shows strong clustering around the ideal line (perfect prediction). The model performs well for typical loads (\$1,000–\$5,000) with increasing variance at extreme prices ($>8,000$), which represent rare premium lanes or peak-season rush shipments.

7.2 Residual Analysis

Residuals (Actual – Predicted) are centered near zero with a mean of \$39.33, indicating **no systematic bias**. The standard deviation of \$677.12 reflects the long-tail distribution of freight rates, where a small number of high-value loads drive variance.

8 Conclusion

This assessment demonstrates a production-ready approach to freight rate prediction:

- Rigorous validation:** Time-based split prevents data leakage and simulates real deployment.
- Domain-aware features:** Geographic, temporal, and freight-specific signals capture market dynamics.
- Strong performance:** 6.87% MAPE is competitive for the freight industry, where rates fluctuate with fuel costs, weather, and demand shocks.
- Actionable insights:** The December weekly cycle proves the model learns *market behavior*, not just distance–price correlation.

Future improvements: Incorporating external data (fuel prices, weather, port congestion) and experimenting with ensemble methods (XGBoost, LightGBM) could further reduce MAPE on high-value outliers.

Repository: github.com/khalilben22/freight-rate-ml-assessment